Haider Riaz Khan

haiderriazkhan.com   haiderriazkhan@gmail.com
haider-riaz-57593aba   haiderriazkhan

Education

2024 – 2026 **M.S.** in Mathematics — City College of New York

2017 – 2018 **M.A.** in Philosophy — University of Waterloo
Thesis: The Phenomenological Origins of Property (Advised by John Turri)

2010 – 2015 **B.Sc.** in Physics and Computer Science — McGill University

Engineering Experience

2025 - Pres **Co-Director and Lead Engineer** — Red CALENDAR (NYC, USA)
Lead engineering across projects, beginning with Red Calendar, an independent NYC arts and action calendar, and a custom events platform for Wesleyan Center for the Arts. Built hybrid lexical and semantic search over event data, submission and editorial workflows, authentication, syndication, analytics, and automated deployment infrastructure.

2024 - Pres **Software Engineer** — BASE Education (NYC, USA)
Wrote GraphQL APIs and built out the application logic for delivering data insights—including computing student-level data, aggregate statistics, and advanced features such as differentially private statistics. Created fast, efficient, and secure data pipelines that power the backend systems.

2022 - 2023 **Lead Software Engineer** — Terrace (NYC, USA)
Integrated multiple centralized digital exchanges into Terrace's liquidity platform using a FIX API solution. Worked with Circle REST API suites to enable global payments, withdrawals, and trades using USDC.

2021 - 2022 **Senior Software Engineer** — Avenue 8 (NYC, USA)
Implemented Avenue 8's observability infrastructure. Wrote automation, monitoring, and alerting scripts. Automated the management of AWS resources using Terraform. Improved network security.

2019 - 2021 **Software Engineer** — Lifion by ADP (NYC, USA)
Built Lifion's graph database framework using a microservices architecture, Node.js, Gremlin, and distributed message queuing services. Wrote optimized Gremlin queries and SQL subroutines.

2016 - 2017 **Software Engineer** — Medidata Solutions (NYC, USA)
Contributed to the development of a Java-based combinatorial test data generator called BIT-TAG.

Awards & Honors

2025	Dr. Barnett and Jean Hollander Rich Graduate Scholarship	City College
2025	Dr. Barnett and Jean Hollander Rich Summer Fellowship	City College
2018	Paul Seligman Memorial Scholarship	University of Waterloo
2015	Dean's Multidisciplinary Undergraduate Research List	McGill
2014	NSERC-CREATE Neuroengineering Award	McGill
2010	Alexander Rutherford Scholarship	Western Canada High School

Publications

Conference and Journal Articles

1. **Khan, Haider Riaz**, & John Turri. "Phenomenological Origins of Psychological Ownership." *Review of General Psychology* 26, no. 4 (December 2022): 446–63.

2. Li, Nan, Yu Lei, **Haider R. Khan**, Jingshu Liu, and Yun Guo. "Applying Combinatorial Test Data Generation to Big Data Applications." In *Proceedings of the 31st IEEE/ACM International Conference on Automated Software Engineering (ASE '16)*, 637-47. New York, NY: Association for Computing Machinery, August 2016.

Essays

1. "The Forever War." *Commune*, Issue 5, Winter 2020.

Research & Teaching

2018 **Translator** — Philosophical Science Lab, University of Waterloo

Translated English stimuli into Pashto and Urdu for the purposes of a cross-cultural study of the "ought implies can" principle. Provided feedback on research paper drafts to Prof. Turri.

Winter 2018 **Teaching Assistant** — University of Waterloo

Biomedical Ethics (Prof. Andrew Stumpf)

Fall 2017 **Teaching Assistant** — University of Waterloo

Professional and Business Ethics (Dr. Jim Jordan)

2014 - 2015 **NSERC Neuroengineering Fellow** — Montreal Neurological Institute

Wrote CANDLE-J, an open-source 3D image denoising software designed as an ImageJ plugin. CANDLE-J is adept at processing deep in vivo 3D multiphoton microscopy images where the signal-to-noise ratio (SNR) is low. It is written in Python, Java, and multithreaded C.

2012 - 2014 **Computational Scientist** — Cook Lab, McGill University

Developed MATLAB routines to measure the cross-correlation of microsaccades and microstimulations (in area MT of the visual cortex). Created a library of MATLAB functions to compute joint metrics of neural activity. The joint metrics are computed for both single-unit and multi-unit neuronal spikes.

Fall 2011 **SUS Peer Tutor** — McGill University

Met with students routinely to go over physics and chemistry concepts; held exam review sessions.

Talks & Presentations

2016 comBinatorial bIg daTa Test dAta Generator (BIT-TAG) — ASE'16, Singapore

2014 Neuroscience From Below — Student Summer Colloquium, McGill

TECHNICAL SKILLS

Programming Languages: JavaScript/TypeScript, Clojure, Java, Python, C/C++, Bash, SQL
Data, Storage & Querying: MySQL/Postgres, pgvector, S3, GraphQL, Elasticsearch, Redis
Development & Cloud: Git, GitHub Actions, Docker, AWS Lambda
Scientific Computing: MATLAB, Jupyter, SageMath, ImageJ, dtype-next

Languages

English, Urdu, Pashto